

cp_2e

1.

Greater than Five - 2 | 🟡 15/15

Easy

Write a program that reads three numbers `A` , `B` , and `C` , and checks if each of the given numbers is greater than `5`.

Input

The first line of input contains an integer representing `A` .

The second line of input contains an integer representing `B` .

The third line of input contains an integer representing `C` .

Output

The output should be a single line containing a boolean. **True** should be printed if each of the given numbers is greater than `5`, otherwise **False** should be printed.

Explanation

For example, if the given numbers are `A = 7` , `B = 18` and `C = 239` ,

✓ `A` is greater than `5`. (`7` is greater than `5`)

✓ `B` is greater than `5`. (`18` is greater than `5`)

✓ `C` is greater than `5`. (`239` is greater than `5`)

The output should be **True** as all the given numbers are greater than `5`.

Sample Input 1

```
7
18
239
```

Sample Output 1

```
True
```

Sample Input 2

```
12
6
1
```

Sample Output 2

```
False
```

code:

```
n1=int(input())
n2=int(input())
n3=int(input())
s=n1>5 and n2>5 and n3>5
print(s)
```

2.

Greater than or Equal to 20 | 🏆 20/20

● Solve

Easy

Write a program that reads three numbers `A` , `B` , and `C` and checks if each number is greater than or equal to `20`.

Input

The first line of input contains an integer representing `A` .
The second line of input contains an integer representing `B` .
The third line of input contains an integer representing `C` .

Output

The output should be a single line containing a boolean. **True** should be printed if each number is greater than or equal to `20`, otherwise **False** should be printed.

Explanation

For example, if the given numbers are `A = 10` , `B = 20` and `C = 30` ,

✗ `10` is greater than or equal to `20`. (`10` is less than `20`)

✓ `20` is greater than or equal to `20`. (`20` is equal to `20`)

✓ `30` is greater than or equal to `20`. (`30` is greater than `20`)

The output should be **False** as not all of the numbers greater than or equal to `20`.

Sample Input 1

```
10
20
30
```

Sample Output 1

```
False
```

Sample Input 2

```
60
45
39
```

Sample Output 2

```
True
```

code:

```
n1=int(input())
n2=int(input())
n3=int(input())
s=n1 >=20 and n2>=20 and n3>=20
print(s)
```

3.

Compare Digits - 2 🟡 25/25

Write a program to check if the given two digit number is greater than 25 and its first digit is greater than its second digit.

Input

The first line of the input will be a two digit integer

Output

Print "True" if the number is greater than 25 and its first digit is greater than its second digit.
In all other cases print "False".

Explanation

When the given number is 24:

- ✗ Greater than 25
- ✗ First digit greater than second digit (2 is less than 4)

When the given number is 42:

- ✓ Greater than 25
- ✓ First digit greater than second digit (4 is greater than 2)

Sample Input 1

24

Sample Output 1

False

Sample Input 2

42

Sample Output 2

True

code:

```
n=input()  
s=int(n)>25  
s1=n[0]>n[1]  
s2=s and s1  
print(s2)
```

4.

Sum and Product Check | 🟡 20/20

● Solved

Easy

Write a program that reads two numbers `A` and `B` and checks if both the sum and the product of the given numbers have less than three digits.

Input

The first line of input contains an integer representing `A` .
The second line of input contains an integer representing `B` .

Output

The output should be a single line containing a boolean. **True** should be printed if both the sum and the product of the given numbers have less than three digits, otherwise **False** should be printed.

Explanation

For example, if the given numbers are `A = 17` and `B = 4` ,

✓ The sum of `A` and `B` has less than three digits. ($17 + 4 = 21$, **21** has less than three digits)

✓ The product of `A` and `B` has less than three digits. ($17 * 4 = 68$, **68** has less than three digits)

The output should be **True** as both the sum and the product of the given numbers have less than three digits.

Sample Input 1

```
17
4
```

Sample Output 1

```
True
```

Sample Input 2

```
12
100
```

Sample Output 2

```
False
```

code:

```
n1=int(input())
n2=int(input())
s=len(str(n1+n2))<3
s2=len(str(n1*n2))<3
print(s and s2)
```

5.

Digit Zero | 🟡 20/20

Easy

Write a program that reads a three-digit number and checks if the given number contains **0**.

Input

The input will be a single line containing a three-digit integer.

Output

The output should be a single line containing a boolean. **True** should be printed if the given number contains **0**, otherwise **False** should be printed.

Explanation

For example, if the given number is **120**,

✗ The first digit is equal to **0**. (1 is not equal to 0)

✗ The second digit is equal to **0**. (2 is not equal to 0)

✓ The third digit is equal to **0**. (0 is equal to 0)

The output should be **True** as the given number contains **0**.

Sample Input 1

120

Sample Output 1

True

Sample Input 2

678

Sample Output 2

False

code:

```
n=input()  
a=int(n[0])==0  
b=int(n[1])==0  
c=int(n[2])==0  
print(a or b or c)
```

6.

Compare with Ten - 2 | 🟡 25/25

● Solve

Easy

Write a program that reads three numbers `A` , `B` , and `C` , and checks if the sum of any two numbers is **always** greater than **10**.

Input

The first line of input contains an integer representing `A` .
The second line of input contains an integer representing `B` .
The third line of input contains an integer representing `C` .

Output

The output should be a single line containing a boolean. **True** should be printed if the sum of any two numbers is **always** greater than **10**, otherwise **False** should be printed.

Explanation

For example, if the given numbers are `A = 4` , `B = 8` and `C = 17` .

The sum of `A` , `B` and the sum of `B` , `C` and the sum of `C` , `A` should be greater than **10**.

✓ `A + B` is greater than **10**. (`4 + 8 = 12` , **12** is greater than **10**)

✓ `B + C` is greater than **10**. (`8 + 17 = 25` , **25** is greater than **10**)

✓ `C + A` is greater than **10**. (`17 + 4 = 21` , **21** is greater than **10**)

The output should be **True** as the sum of any two numbers is always greater than **10**.

Sample Input 1

```
4
8
17
```

Sample Output 1

```
True
```

Sample Input 2

```
121
3
5
```

Sample Output 2

```
False
```

code:

```
a=int(input())
b=int(input())
c=int(input())
n1=(a+b)>10
n2=(a+c)>10
n3=(b+c)>10
print(n1 and n2 and n3)
```

7.

Easy

Write a program that reads a three-digit number and checks if each digit is greater than 4 or the first digit is equal to 6.

Input

The input will be a single line containing a three-digit integer.

Output

The output should be a single line containing a boolean. **True** should be printed if each digit is greater than 4 or the first digit is equal to 6, otherwise **False** should be printed.

Explanation

For example, if the given number is **612**,

✗ Each digit is greater than 4. (All the digits should be greater than 4 but 1 and 2 are not greater than 4)

✓ The first digit is equal to 6. (The first digit of **612** is equal to 6)

The output should be **True** as the first digit of the given number **612** is equal to 6.

Sample Input 1

612

Sample Output 1

True

Sample Input 2

123

Sample Output 2

False

code:

```
n=input()  
n1=int(n[0])>4  
n2=int(n[1])>4  
n3=int(n[2])>4  
n4=int(n[0])==6  
print((n1 or n2 or n3) or n4)
```

8.

Easy

Write a program that reads a three-digit number and checks if all the below conditions are satisfied.

- The number contains **1**.
- The sum of all the digits of the number is less than **12**.
- The last digit of the number is equal to **5**.

Input

The input will be a single line containing a three-digit integer.

Output

The output should be a single line containing a boolean. **True** should be printed if all the given conditions are satisfied, otherwise **False** should be printed.

Explanation

For example, if the given number is **315**,

- ✓ The number contains **1**. (The second digit of **315** is equal to **1**)
- ✓ Sum of all the digits of the number is less than **12**. ($3 + 1 + 5 = 9$, **9** is less than **12**)
- ✓ The last digit of the number is **5**. (The last digit of **315** is **5**)

The output should be **True** as all the given conditions are satisfied.

Sample Input 1

315

Sample Output 1

True

Sample Input 2

241

Sample Output 2

False

code:

```
n=input()  
n1=int(n[0])==1  
n2=int(n[1])==1  
n3=int(n[2])==1  
sum_=(int(n[1])+int(n[0])+int(n[2]))<12  
last_dig=int(n[2])==5  
print((n1 or n2 or n3)and sum_ and last_dig)
```

9.

Compare Numbers - 9 | 🟡 30/30

● Solved

Easy

Write a program that reads a three-digit number and checks if any of the below conditions is satisfied.

- Each digit of the given number is greater than 7.
- The product of any two digits is **always** less than or equal to 30.

Input

The input will be a single line containing a three-digit integer.

Output

The output should be a single line containing a boolean. **True** should be printed if each digit of the given number is greater than 7 or the product of any two digits is always less than or equal to 30, otherwise **False** should be printed.

Explanation

For example, if the given number is 832,

- ▶ ✗ Each digit is greater than 7.
- ▶ ✓ The product of any two digits is **always** less than or equal to 30.

The output should be **True** as the product of any two digits is **always** less than or equal to 30.

Sample Input 1

832

Sample Output 1

True

Sample Input 2

649

Sample Output 2

False

code:

```
n=input()
n1=int(n[0])>7
n2=int(n[1])>7
n3=int(n[2])>7
pr=(int(n[0])*int(n[1]))<=30
pr1=(int(n[1])*int(n[2]))<=30
pr2=(int(n[2])*int(n[0]))<=30
c1=n1 and n2 and n3
c2=pr2 and pr1 and pr
print(c1 or c2)
```

10.

Eligibility Criteria - 2 | 🏆 25/25

Medium

Write a program that reads the marks in Maths M , Physics P , and Chemistry C , and checks if all the below conditions are satisfied.

- $M + P \geq 100$ or $P + C \geq 100$ or $M + C \geq 100$
- $M + P + C \geq 180$

Input

The first line of input contains an integer representing M .

The second line of input contains an integer representing P .

The third line of input contains an integer representing C .

Output

The output should be a single line containing a boolean. **True** should be printed if all the given conditions are satisfied, otherwise **False** should be printed.

Explanation

For example, if the given marks are $M = 82$, $P = 55$, and $C = 45$,

✓ $M + P \geq 100$ or $P + C \geq 100$ or $M + C \geq 100$. ($82 + 55 = 137$, **137** is greater than or equal to **100**)

✓ $M + P + C \geq 180$. ($82 + 55 + 45 = 182$, **182** is greater than **180**)

The output should be **True** as all the given conditions are satisfied.

Sample Input 1

```
82
55
45
```

Sample Output 1

```
True
```

Sample Input 2

```
71
30
70
```

Sample Output 2

```
False
```

code:

```
m=int(input())
p=int(input())
c=int(input())
s1=(m+p)>=100
s2=(p+c)>=100
s3=(c+m)>=100
s4=(m+p+c)>=180
print((s1 or s2 or s3) and s4)
```